

Lab 3: Component Modelling & Architectural Pattern Selection

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1. Architecture Selection

Architecture Selection Statement:

"We chose Layered Architecture for the Digital Gift Card & Loyalty Point Exchange System."

- Selected Pattern: Layered Architecture
- Reason 1 (Scenario-Related): The system has clear separation between user interaction and core business operations. The Presentation Layer handles Account Holder interactions, while the Business Layer manages loyalty-point validation, exchange-credit conversion, gift-card redemption, voucher locking, payment processing, and merchant management. This separation makes the system easier to understand, maintain, and modify.
- Reason 2 (Scenario-Related): NFR-001 requires point conversion and gift-card redemption operations to complete within 250 ms with zero ledger balance drift. Keeping the core exchange and redemption operations within a dedicated Business Layer allows these critical operations to be coordinated consistently by the Order Manager while keeping presentation concerns separate.
- Security Advantage: Layered separation ensures that sensitive operations such as payment processing, exchange-credit handling, and anti-fraud voucher locking are controlled through the Business Layer rather than being accessed directly from the user interface. This supports the security requirements of PCI-DSS, AES-256 encryption at rest, TLS 1.3 in transit, and immutable audit logging specified in NFR-002.
- Performance Benefit: Separating the Presentation Layer from the Business Layer reduces unnecessary processing in the user-interface component. The Order Manager can directly coordinate the loyalty/exchange, payment,

and voucher operations, helping maintain the required low latency for point conversion and gift-card redemption.

2. Identified System Components & Interfaces

A. Components

The system contains 6 components:

User Interface Component: Provides the Account Holder-facing interface for wallet interaction, gift-card browsing, redemption confirmation, and displaying exchange rates and resulting exchange credits.

1. Order Manager Component: Acts as the central workflow coordinator. It receives requests from the User Interface Component and coordinates loyalty/exchange operations, payment processing, and gift-card/voucher redemption.
2. Payment Service Component: Handles payment processing requests from the Order Manager Component.
3. Loyalty & Exchange Component: Validates merchant loyalty points, aggregates loyalty points, applies dynamic exchange rates, converts loyalty points into universal exchange credits, and manages exchange-credit balance operations.
4. Gift Card & Voucher Component: Handles gift-card redemption, voucher generation, cryptographic checksums, voucher locking, and anti-fraud protection against duplicate redemption.
5. Merchant Management Component: Manages merchant gift-card catalogs, publishes loyalty programs, and provides gift-card redemption reports for Merchant Partners.

B. Key Interfaces

The diagram contains 5 interfaces:

6. Order Interface: Connects the User Interface Component with the Order Manager Component. It handles Account Holder order and redemption requests.

7. Payment Interface: Connects the Order Manager Component with the Payment Service Component. It handles payment processing requests.
 8. Loyalty / Exchange Interface: Connects the Order Manager Component with the Loyalty & Exchange Component. It supports loyalty-point validation, exchange-rate application, balance validation, and conversion to universal exchange credits.
 9. Voucher Interface: Connects the Order Manager Component with the Gift Card & Voucher Component. It supports gift-card redemption, voucher generation, voucher locking, and anti-fraud operations.
 10. Merchant Interface: Connects the Gift Card & Voucher Component with the Merchant Management Component. It supports access to merchant gift-card catalog and merchant-related information required during the redemption workflow.
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3. UML Component Diagram

